Comprehensive Project Plan: Bundestag Roll-Call Scaling (Option B)

This document outlines the complete project plan for analyzing the 20th German Bundestag roll-call data, specifically tailored for **Option B (Automated Coding via AI)**. It integrates the core analytical tasks with the necessary infrastructure, presentation, and transparency requirements.

Phase 1: Analytical Work Packages

Work Package 1: Data Pre-processing & Imputation

Objective: Transform the raw voting records into a complete matrix. Matrix factorization methods require complete data.

- **Task 1.1: Data Integration.** Load bundestag_wp132_polls.csv and bundestag_wp132_votes.csv. Recode votes: "ja" = 1, "nein" = 0, and all others = NA.
- **Task 1.2: Matrix Construction.** Reshape the data into a rectangular matrix (rows = individual legislators, columns = roll-call votes).
- **Task 1.3: Double-Mean Imputation.** Fill missing values using the double-mean formula. This intuitively balances an individual MP's baseline tendency to agree with the overall popularity of a specific bill.

Work Package 2: Baseline Singular Value Decomposition (SVD)

Objective: Extract the primary ideological dimension of the parliament using geometric reduction.

- **Task 2.1: Execute Truncated SVD.** Factorize the imputed matrix to identify the latent coordinates of legislators (rows) and bills (columns).
- **Task 2.2: Political Interpretation.** Analyze the first dimension. Identify the legislators and votes at the extreme ends. Assess if the distribution aligns with the German government/opposition divide.
- **Task 2.3: Second Dimension Analysis.** Evaluate the second extracted dimension to see if it represents a meaningful secondary political conflict or merely statistical noise.
- **Task 2.4: Variance Calculation.** Compute the proportion of total variance explained by the first dimension using the singular values.

Work Package 3: Advanced Geometric Pre-processing (Double-Centering)

Objective: Strip away baseline popularities (e.g., unanimous bills or always-agreeing MPs) to focus strictly on polarization.

- **Task 3.1: Apply Double-Centering.** Subtract the row mean and column mean from each matrix element, then add the grand mean. Verify that rows and columns sum to zero.
- **Task 3.2: Re-run SVD.** Perform the SVD analysis on this new matrix.
- **Task 3.3: Comparative Evaluation.** Document how the ordering of legislators changes when baseline effects are removed.

Work Package 4: Probabilistic Modeling (IRT)

Objective: Model voting as a probabilistic outcome with uncertainty, rather than rigid geometric points.

- **Task 4.1: Model Setup.** Define a 1-Dimensional 2-PL IRT model using brms or a Python equivalent. Ensure discrimination parameters can be negative to capture both left- and right-wing polarization.
- **Task 4.2: Prior Definition.** Set standard informative priors (e.g., standard normal for ideal points) and apply sign constraints to anchor the ideological spectrum.
- **Task 4.3: Model Fitting.** Estimate the model via MCMC and extract the posterior distributions for the MPs' ideal points.
- **Task 4.4: Cross-Method Comparison.** Compare the Bayesian ideal points with the SVD coordinates from WP 2 and WP 3.

Work Package 5: Substantive Evaluation

Objective: Utilize the Bayesian posterior distribution to test a real-world political claim.

- **Task 5.1: Formulate Claim.** Choose a specific claim about the 20th Bundestag's ideological landscape.
- **Task 5.2: Individual Uncertainty.** Use Credible Intervals (CIs) to demonstrate how precisely an individual MP's position is known.
- **Task 5.3: Ordinal Uncertainty.** Calculate the pairwise ordering probability across MCMC samples to confidently evaluate if one actor is statistically positioned differently than another.

Phase 2: Option B Presentation & Infrastructure (AI Track)

Work Package 6: Infrastructure & Repository Setup

Objective: Establish the technical foundation for the project submission.

- **Task 6.1: Version Control.** Create a public Git repository (e.g., GitHub, GitLab).
- **Task 6.2: Hosting Configuration.** Set up GitHub Pages or Netlify to host the repository as a static, publicly accessible website.
- **Task 6.3: Project Structure.** Organize the repository logically with dedicated folders for raw data, processed data, scripts, visualizations, and web assets.

Work Package 7: Narrative Structure & Integration

Objective: Combine analytical rigor with readable, engaging web content.

- **Task 7.1: Seamless Embedding.** Integrate the explanatory text seamlessly with the corresponding code snippets and model outputs (Steps 1-4).
- **Task 7.2: Intuitive Explanations.** Explain complex concepts like Bayesian posterior distributions clearly and accessibly.

Work Package 8: Visualization & Corporate Design

Objective: Ensure all visual elements are professional, clear, and on-brand.

- **Task 8.1: University Branding.** Apply the University of Mannheim color palette (e.g., navy blue, ice blue) to the website layout and all generated plots.
- **Task 8.2: Clear Labeling.** Ensure all figures use properly formatted MP names rather than raw dataset IDs.

Work Package 9: The Interactive Component

Objective: Provide an engaging element that allows the reviewer to explore the results.

- **Task 9.1: Interactive Implementation.** Use libraries like Plotly or htmlwidgets to create at least one interactive visualization (e.g., a scatterplot with tooltips showing MP names and parties on hover).
- **Task 9.2: Functionality Testing.** Verify that the interactive element works smoothly on

the live hosted webpage.

Work Package 10: Documentation of AI Interactions

Objective: Maintain transparency regarding the use of AI in generating the code and analysis.

- **Task 10.1: Export Chat History.** Compile the complete history of prompts and AI responses used during the project.
- **Task 10.2: Public Accessibility.** Include this history as a clearly labeled file (e.g., AI_Prompts.md) in the main directory of the public repository.